Calculus II

Problems for Training #2 24–28 Feb 2025

◇ **Topics:**

- Limits (continued)
- Limit along paths
- Continuity

Problem 1. By considering different paths of approach, show that the functions have no limit as $(x, y) \to (0, 0)$.

$$f(x, y) = \frac{x^4}{x^4 + y^2}.$$

In Problems 2–3, investigate at what points (x, y) in the plane the functions are continuous.

Problem 2.

$$g(x, y) = \sin \frac{1}{xy}$$

Problem 3.

$$g(x, y) = \frac{x + y}{2 + \cos x}$$

In Problems 4–7, investigate at what points (x, y, z) in space the functions are continuous.

Problem 4.

$$f(x, y, z) = x^2 + y^2 - 2z^2$$

Problem 5.

$$f(x, y, z) = \sqrt{x^2 + y^2 - 1}$$

Problem 6.

$$h(x, y, z) = \ln(z - x^2 - y^2 - 1)$$

Problem 7.

$$h(x, y, z) = \frac{1}{z - \sqrt{x^2 + y^2}}$$

For Problems 8–10, let

$$f(x,y) = \begin{cases} x^2, & x \geq 0 \\ x^3, & x < 0. \end{cases}$$

Find the following limits.

Problem 8.

$$\lim_{(x,y)\to(3,-2)} f(x,y).$$

Problem 9.

$$\lim_{(x,y)\to(-2,1)} f(x,y).$$

Problem 10.

$$\lim_{(x,y)\to(0,0)} f(x,y).$$

END

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Answers to selected problems

3. $\mathbb{R}^2$
4. $\mathbb{R}^3$
5. $\{(x,y,z) \in \mathbb{R}^3 : \ x^2 + y^2 \geq 1\}$
6. $\{(x,y,z)\mathbb{R}^3 : \ z - x^2 - y^2 - 1 > 0\}$
7. $\{(x,y,z) \in \mathbb{R}^3 : \ z - \sqrt{x^2+y^2} \neq 0\}$
8. 9
9. -8
10. 0